Columbia University Consent Form

Protocol Information

Attached to Protocol: IRB-AAAK6753
Principal Investigator: Julie Herbstman (jh2678)
IRB Protocol Title: Fair Start Study

General Information

Consent Number: CF-FRYX9700
Participation Duration: 18 years
Anticipated Number of Subjects: 800

Research Purpose: You are invited to participate in a research study to understand how environmental exposures during pregnancy influence the health and development of children.

This consent form is written to address a research subject. If, however, you will be providing permission as the parent or legal guardian of a minor [OR] a legally authorized representative, the words 'you' and 'your' should be read as 'your child' [OR] 'the research subject'.

Contacts

Contact	Title	Contact Information
Catherine Tobon	Coordinator	Phone: 347-266-9132 Email: ct2405@cumc.columbia.edu
Julie Herbstman	Principal Investigator	Phone: 917-575-2884 Email: jh2678@cumc.columbia.edu
Maria de Jesus Hernandez Garcia	Other Engaged Personnel	Phone: 646-531-7389 Email: mh4268@cumc.columbia.edu
Laura Mejia Pinilla	Coordinator	Phone: 646-531-5421 Email: lmm2317@cumc.columbia.edu
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Contact	Title	Contact Information
Camila Gaitan	Other Engaged Personnel	Phone: 929-777-1154 Email: cg3433@cumc.columbia.edu

Information on Research

You and/or your child may be able to complete visits on a computer or tablet, by mail or email, over the phone, by text message or in person. At each visit, study staff will contact you to explain the purpose of the study, the procedures to be used, and the possible benefits and risks of your and your children's participation. Please ask the study staff to explain anything that you do not understand.

This study looks at how contaminants in your environment and your child's environment may affect your child's health. The study may help identify health concerns such as breathing difficulties, lead poisoning, developmental delays, or being overweight. If needed, we will help you identify special care or services that may help your child grow up healthier.

Some study results may help us understand your child's health. Other results will help us understand patterns among all the children in the study. The wristband, blood, and urine samples may not tell us much about your child's individual health. However, when we look at results from hundreds of mothers and children together, we can answer important questions about how pollution may affect children's health. A trained research worker will work with you and your child, ask you questions, conduct developmental evaluations, take measurements, and collect samples as described below. Some of the samples and questionnaires may be collected during the same visit as your ECHO study visit.

If you agree to participate, the following will happen at each visit:

At recruitment, a research worker will ask you questions about yourself and your family.

PRENATAL VISIT

During your first or second trimester a research worker will:

- Ask you for a urine sample.
- Ask you to take some body measurements.
- Ask you for a blood sample from your arm (about 25-35 mL, which is equal to 5-7 teaspoons).

During your third trimester of pregnancy, a research worker will:

- Mail you a collection package to the address we have on file, and a research worker will be available to answer questions or concerns.
- Ask you to wear a small plastic wristband for 48 hours. We will send the wristband and a Limited Data Set (LDS) to our collaborator, Dr. K. Anderson, at Oregon State University. An LDS excludes certain information that could be used to identify you. The collaborator will use the wristband and LDS to measure chemicals you may have been exposed to.

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- Interview you about your pregnancy, including your eating and drinking habits, we will ask a number of questions about your home and neighborhood, and how you prepare your food, activities, medical history, and how you have been feeling during the past year.

When we collect the wristband, we will ask you another series of questions about where you went, your eating habits, and other activities that you did during those 48 hours only.

- Ask you for several urine samples during your prenatal visit, after 24 hours, and after 48 hours.
- Ask you for a blood sample from your arm (about 25-35 mL, which is equal to 5-7 teaspoons).
- Ask you to measure your weight, height, waist circumference and blood pressure.

Remote visit:

During your third trimester of pregnancy, a research worker will:

- Ask you to wear a small plastic wristband for 48 hours. The wristband and Limited Data Set will be sent to our collaborator as described above.
- Interview you about your pregnancy using the same topics described above.
- Ask you for several urine samples during your prenatal visit, after 24 hours, and after 48 hours.

DELIVERY

When you deliver your baby at the hospital, a research worker will:

- Collect a blood sample from your baby's umbilical cord (about 30-60 mL, which is equal to 6-12 teaspoons) and a sample of placental tissue before it is discarded by medical staff.
- Ask you for a blood sample from your arm (about 25-35 mL, which is equal to 5-7 teaspoons).
- Collect a sample of your baby's first bowel movement (meconium) from your baby's diaper.

1-5 MONTHS VISIT

When your baby is between 1 and 5 months old, a research worker will ask you to visit our office to:

- Interview you (ask you questions).
- Ask you to provide biological samples from you and your baby such as urine, stool, blood spots.
- Measure your baby's weight, length, and head circumference.

6-11 MONTHS VISIT

When your baby is between 6 and 11 months old, a research worker will ask you to visit our office to:

- Interview you (ask you questions).
- Take some body measurements.
- Measure your baby's weight, length, and head circumference.
- Ask you for a urine sample.

12-23 MONTHS VISIT

When your baby is between 12 and 23 months old, a research worker will ask you to visit our office to:

- Interview you (ask you questions), including questions about your child's behavior.
- Take some body measurements.
- Measure your baby's weight, length, and head circumference.
- Ask you to provide biological samples from you and your baby such as nasal swab, blood spots.

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- Test your baby's development. We will refer you for an early intervention evaluation if needed.

24-35 MONTHS VISIT
When your baby is between 24 and 35 months old, a research worker will ask you to visit our office to:

- Interview you (ask you questions), including questions about your child's behavior.
- Take some body measurements.
- Measure your baby's weight, length, and head circumference.
- Ask you to provide biological samples from you and your baby such as blood spots.
- Test your baby's development. We will refer you for an early intervention evaluation if needed.

3 YEAR VISIT
When your child is 3 years old, a research worker will ask you to visit our office to:

- Interview you (ask you questions), including questions about your child's behavior.
- Take some body measurements.
- Measure your child's weight, length, and head circumference.
- Ask your child for a urine sample.
- Ask you to allow us to take a small blood sample from your child's arm (about 10-15 mL, which is equal to 2-3 teaspoons).
- Test your child's development. We will refer you to services if needed.

When your child is age 4 and older, a research worker will ask you to visit our office, every year to:

Interview you (ask you questions), including questions about your child's behavior.

- Take some body measurements from you, such as height, weight and blood pressure
- Take some body measurements from your child, such as height and weight.
- Ask you to allow us to take a urine sample from your child.
- Ask you to allow us to take a small blood sample from your child's arm (about 15-20 mL, which is equal to 2-3 teaspoons) that we will use to measure different exposures.
- Test your child's development. We will refer you to services if needed.

Your child's blood may be used for different types of tests. Some tests may look for contaminants or other exposures that could affect your child's health. Other tests may examine DNA and genetic factors. More information is provided in the Genetic Information section.

We may also call you when your child is 18, 30, 42, and 54 months old. During these short interviews, we will ask the asthma questionnaire and ask about new symptoms, medical problems, and medications since the last visit.

Risks

There are no expected risks to you or your child in participating in this study. However, the risks of having blood drawn include soreness, bruising where the needle is inserted, and discomfort during the procedure. The amount of blood collected is not expected to pose a significant risk to you or your child.

A risk of taking part in this study is the possibility of a loss of confidentiality or privacy. This means that personal information about you or your child could be seen by someone who is not supposed to see it. The steps we will take to

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protect your information are described in the Confidentiality section.

Benefits

Being in this study may benefit you or your child, but direct benefit is not guaranteed. You will be able to learn about your child's developmental test results after applicable visits. The Columbia Center for Children's Environmental Health has a child psychologist and trained staff who may identify developmental delays during testing and make referrals if needed. Referrals may be made to partner clinics for mental health or other health concerns.

If anything abnormal shows up in your child's blood tests, we will notify you. This information may help you protect your child's health. The information collected may also help future generations.

Alternative Procedures

The alternative is not to participate in this research study.

Confidentiality

Electronic consent will be completed through REDCap. Information collected during the consent process will be stored in a secure REDCap database, and a copy will be saved on a secure server at the Data Coordinating Center at NYSPI (New York State Psychiatric Institute).

We will make every reasonable effort to keep your and your child's information private, but total privacy cannot be guaranteed. Study materials and samples will be labeled with a code number instead of names. The file linking names to code numbers will be stored securely, and only the investigator and authorized study staff will have access.

To help protect your privacy, the study team has a Certificate of Confidentiality from the United States Department of Health and Human Services. This Certificate allows the researchers to refuse to disclose information that may identify you in legal proceedings. It does not prevent required reporting of child abuse or neglect, actions needed to prevent harm, disclosures that you authorize, information you or your family voluntarily release, or federal audits and evaluations.

The following people and agencies may look at, copy, use, or share your research records: the investigator, study staff, and other professionals evaluating the study; Columbia University and NewYork-Presbyterian Hospital, including the Institutional Review Board; the National Institutes of Health; the U.S. Food and Drug Administration; and the Office for Human Research Protections. They are required to protect the confidentiality of your information.

To be in this study, you must give us permission to use and share protected health information and other identifiable information collected for the research.

The health information that may be used and shared includes all information collected during this study and information in your medical records that is relevant to the research, including information about your current pregnancy.

Columbia University Medical Center and NewYork-Presbyterian Hospital may provide relevant health information to the researchers. The researchers may use or share it with NIH and its agents or contractors to conduct the study,

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study the results, and make sure the research was done correctly. Coded address information may be shared with other researchers to estimate pollution levels near your home.

If study results are published or presented, your name, your child's name, and other direct identifiers will not be used. If phone calls or mail are unsuccessful, authorized study staff may review medical records to confirm or update your contact information.

The study team may contact you by phone, text message, email, or direct mail regarding study visits, questionnaires, reminders, study-related information, and future research opportunities. If we are not able to reach you, we may also contact alternate contacts that you have provided or attempt to obtain updated contact information through online search platforms. We use REDCap to manage study information and study-related communications. Text messages sent from REDCap are delivered through Twilio, a text-messaging service, and may include study reminders, survey links, and other study information. The messages will not contain personally identifiable or sensitive study information. Only your phone number is transmitted through Twilio to deliver the message. You may ask us to stop contacting you through a particular method at any time without affecting your participation in the study.

You may change your mind and revoke (take back) this consent and HIPAA authorization at any time and for any reason. To revoke it, contact the Principal Investigator, Julie Herbstman, at 917-575-2884.

If you revoke your authorization, you cannot continue in the study. The researchers may continue to use and share information already collected as allowed by this consent form, but no new identifiable health information will be collected after the date you revoke.

This authorization does not expire automatically.

Compensation

You will be paid after each remote or in-person visit through a TruCentive electronic gift card. You may choose to receive the gift card by email, SMS text message, or both. TruCentive lets you choose from a list of pre-approved businesses.

This payment is not in addition to the ECHO payment.

Visit	**In Person**	
1st /2nd Trimester Prenatal Visit	$150	
3rd Trimester Prenatal Visit	$150	
Delivery	$50	
		Remote (questionnaire-only visits)
1-5 and 6-11 month visits	$75	$40
1-4 year visits	$150	$75
5 years and older visits	$200	$100

Your child will receive a small gift, such as a coloring book, at each in-person developmental visit.

For certain types of compensation, such as Venmo or PayPal, information about you, including bank account, physical

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address, or credit card information, may need to be entered into a secure third-party system. The research team will not have access to the information you enter for compensation purposes.

Additional Costs

There are no costs to you for participating in this study.

Voluntary Participation

Voluntary Participation

Taking part in this study is your choice. You may decide not to participate or may stop at any time. Your decision will not affect your future care or your status with the investigator.

If you stop participating, the researchers may continue to use information already collected as described in this consent form.

You will receive a copy of the signed and dated consent form. By signing, you are not giving up any legal rights.

Sub-Studies Participation

You may be contacted about future sub-studies. You are free to say no.

Each sub-study will have its own consent form. Study staff will explain it to you before you decide whether to participate.

Additional Information

Genetic Information

Your blood sample may be used for different types of tests. Some tests may look for chemicals or other substances in your blood that could affect your health. Other tests may look at genetic factors and examine your DNA.
DNA is the genetic material in your cells that acts like an instruction guide for how your body works. These tests can help researchers understand how the environment and other factors may influence health.
Genetic factors are traits you can inherit from your parents, and they can run in families. If you join this study, we will collect a sample from you and look at your genetic information. To protect your privacy:

- Your sample will be labeled with a special code number instead of your name. When it is sent for analysis or placed in storage, it will not include your name or other direct identifiers.

- Your sample will be studied together with many other people's samples, and the results will be grouped together so that no one can tell which results belong to you.

- You will not get your individual genetic results because researchers do not know what the results would mean for your health.

You should also know that insurance companies sometimes use information from genetic testing to deny coverage to

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applicants. The risk of this affecting you is low because this is genetic research, not a clinical genetic test. Therefore, if you agree to participate and are asked whether you have had genetic testing, you should answer no, since your participation in this genetic research is not the same as genetic testing.

Contact Information

If you have any questions or concerns about the study, you may contact Julie Herbstman at 917-575-2884.

If you have any questions about your rights as a research subject, you may contact: Institutional Review Board
Columbia University Medical Center
154 Haven Ave, 2nd Floor
New York, NY 10032
Telephone: (212) 305-5883
Email: irboffice@columbia.edu

An Institutional Review Board is a committee organized to protect the rights and welfare of human subjects involved in research.

Statement of Consent

I have read this consent form and HIPAA authorization. The research study has been explained to me. I agree to take part in the study and give permission to use and share my health information as described in this form. My questions were answered to my satisfaction. I will receive a copy of this form after I sign it. By signing, I am not giving up any legal rights.

Signatures

Participant Signature Lines

Study Participant

Print Name _____ Signature _____

Date & Time _____

Research Signature Lines

Person Obtaining Consent

Print Name _____ Signature _____

Date & Time _____

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